NA 570 Ammonia Retrofit Project Step #2

Report due Wednesday, April 15th

Presentation due April 13th

1.0 Design Optimization Problem

Building off the initial ammonia retrofit model developed as part of Project Step #1, utilize MATLAB's particle swarm optimizer (*particleswarm*) and the same independent variables:

- E_D: Endurance using ammonia (nm)
- W_C: Cargo deadweight (tons)
- V: Speed (kt)
- Tank type/location (two discrete values: on-deck or in-hold)

You will optimize the cost function (Output #5 from Step #1) subject to the following initial constraints:

- The on-deck tank's volume (Output #1) must be less than 6,000m^3 to fit on the aft deck area.
- The in-hold tank's volume (Output #1) must be less than the volume of the aft hold (22m x 28m x 38m = 23,408)
- The added bending moment (Output #2) should be less than 30% of the still water bending moment (SWBM=4,000,000 kN-m) to avoid major structural work
- The magnitude of the added weight (Output #3), either positive or negative, should not exceed 2.5% of the initial displacement ($\Delta = 200{,}000t$) to ensure the current load lines and scantling draft is sufficient.
- GM (Output #4) should be greater than 2m and less than 8m to ensure sufficient stability and seakeeping characteristics.

Carefully consider how you want to implement these constraints, and others that may make the model converge faster. If using an external penalty function approach, be sure to normalize the individual constraints. Note that you will need to develop two separate optimization problems (on-deck and in-hold) and then compare the results of the two. Study the *particleswarm* MATLAB function options and adjust default settings to see if they change the output (e.g., *InertiaRange*, *MinNeighborsFraction*, *SelfAdjustmentWeight*, *SocialAdjustmentWeight*, *SwarmSize*).

Based on your analysis and conclusions from Step #1, modify your baseline optimization problem developed above (includes Step #1 model formulation and initial constraints) and analyze the results based on the following:

- Select <u>two</u> aspects of the reduction of the problem and modify the model to reflect this (e.g., an oversimplification, something not needed, something missing). Rerun your optimization and compare the results to the baseline case.
- Select <u>two</u> technical constraint that you made a different recommendation for and implement it in the optimization problem instead of the initial constraint provided above. Rerun your optimization and compare the results to the baseline case.

2.0 Initial Concept Design and HAZID

Based on the optimization results, select what you believe is the best option, and develop an initial concept design. Propose an initial general arrangement (in PowerPoint or CAD) that includes at least the following systems/spaces in addition to the fixed spaces (i.e., engine room, deckhouse and cargo holds):

- Ammonia fuel storage tank + tank connection space
- Fuel preparation room
- Bunker station
- Vent mast

Briefly describe each system/space above and important characteristics (e.g., fuel storage pressure and temperature) in a dedicated document.

Once the general arrangement and system descriptions are complete, conduct a mini-HAZID workshop for your initial concept design as a team to identify the main hazards for the four nodes above, potential causes and consequences, hazard category, complete a risk rating, and describe effective safeguards and recommendations to reduce risk. Identify and complete an assessment of at least three hazards per node above (4 nodes x 3 hazards = 12 total hazards minimum). A sample risk register template is provided with this assignment. Present the final risk matrix to summarize your risk ratings.

Compile all your recommendations into a short Risk Mitigation Action Plan that you as a team would follow to improve the concept design to reduce identified risks. You do **NOT** need to update your concept design or execute the action plan.

3.0 Report

In a short memo (2-3 pages), address the following:

- Describe how you developed the optimization problem including selection of parameters/options, variable ranges, and implementation of the constraints.
- Present the results of the optimization for the baseline case and the four modified cases (different reductions of problem, different constraints).
- Give your recommendation for the case that should be used (base case or one of the modified ones) and the final optimal variable values and associated cost. Is there one clear winner or are there multiple good candidates?
- Take a step back and consider the original request (see section below on the design problem and background). Did you fulfill this request? How would you approach it differently if you were able to set up the model from scratch? Would you select different variables, outputs, or objectives? Do the results of your analysis make sense from a practical engineering and business perspective?
- Discuss the next steps you could take in this analysis to go into more detail? Can you narrow the design space? Do some disciplines need to have more detailed models?
- Present a figure and describe your initial concept design including the locations of the identified systems/spaces.
- Summarize the main hazards identified in the mini-HAZID and highlight your top risks and how you would plan to reduce them as part of next phase of the project.

4.0 Presentation

Develop a set of presentation slides that your group will use to present the overall project to class. The presentation should be 10 minutes in length with an additional 5 minutes reserved for questions (15 minutes total per group). The presentation should include an overview of the project (Steps #1 and #2), analysis completed, responses to the questions addressed in your two memos (Step #1 and #2), and your main conclusions. The presentation should also include additional visuals to help communicate and present your findings from each step (a picture is worth a thousand words). Prepare your presentation based on the developed slides ensuring that each group member has a part to present and that you maintain the 10-minute length allocated.